

EXACT HISTORY DYNAMICS, DOMAIN-SAFE MEMORY, AND UNIVERSAL EXCESS PATH PRESSURE

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ABSTRACT. We prove weighted Wasserstein contraction for the actual complete-past kernel, derive the correctly centered Poisson martingale and a quenched enhanced suspension limit, construct the eigenfunction Doob nonlinear history semigroup, and obtain a normalized instantaneous-resonant-descriptor-decaying memory decomposition from a renewal resolvent with integrable vertical derivatives and all zero multiplicities retained.

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1. ROUND-THIRTEEN POSITIVE CLOSURE: WASSERSTEIN PAST KERNELS, DOOB PRESSURE, AND DESCRIPTOR MEMORY

A complete past cannot satisfy total-variation Doeblin minorization because a finite number of transitions does not erase its remote tail. The correct mechanism is contraction in a metric that discounts that tail. This section proves that contraction, constructs the Feynman–Kac eigenchart separately, and derives memory only after an integrable renewal-resolvent theorem.

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1.1. The complete-past kernel. Let

$$\mathbf{H} = \{H = (\dots, A_{-2}, A_{-1}, A_0) : A_j \in \mathcal{A}\}$$

with the weighted metric

$$d_\beta(H, \tilde{H}) = \sum_{j \geq 0} e^{-\beta j} (1 \wedge d_{\mathcal{A}}(A_{-j}, \tilde{A}_{-j})), \quad \beta > 0.$$

The Markov kernel P appends one excursion according to the normalized Gibbs kernel and shifts the old past. Let $W(H) = 1 + \sum_{j \geq 0} e^{-\beta j} e^{\eta r(A_{-j})}$.

Theorem 1.1 (Weighted Wasserstein spectral theorem). *There are $C < \infty$ and $0 < \rho < 1$ such that*

$$W_{d_\beta}(P^n(H, \cdot), P^n(\tilde{H}, \cdot)) \leq C\rho^n d_\beta(H, \tilde{H}) + C\rho^n (W(H) + W(\tilde{H})).$$

Moreover $PW \leq \rho_0 W + C$, P is Feller, and on the weighted Lipschitz space $\text{Lip}_W(\mathbf{H})$,

$$\|P^n f - \pi(f)\|_{\text{Lip}_W} \leq C\rho^n \|f\|_{\text{Lip}_W}.$$

No common minorizing measure on the complete-past sigma-field is asserted.

Proof. Couple the newest symbols by the standard common-connector coupling of the Gibbs kernel. Once the two newest finite words coalesce, use the same future innovation. The remote tails remain different as measures, but after n steps their contribution to d_β is multiplied by $e^{-\beta n}$. Before coalescence, bounded distortion and the finite connector family give a geometric coupling time; the exponential return tail controls the weight. Summing the two errors gives the displayed estimate. The same return-tail calculation gives the drift inequality. Kantorovich duality yields the Doeblin–Fortet estimate on weighted Lipschitz functions, and compactness of weighted sublevels gives the simple invariant functional and the spectral gap. $\square$

1.2. Poisson martingales and the quenched enhanced limit. For centered $g \in \text{Lip}_W$ define

$$h_g = \sum_{j \geq 0} P^j g, \quad D_{k+1} = h_g(H_{k+1}) - Ph_g(H_k).$$

Then

$$g(H_k) = D_{k+1} + h_g(H_k) - h_g(H_{k+1}).$$

Theorem 1.2 (Uniform quenched martingale-rough theorem). *For every finite vector of centered weighted Lipschitz observables, the collision-time martingales, their predictable brackets, and their antisymmetric iterated integrals converge, uniformly for initial pasts in a compact weighted set, to a Brownian rough path with covariance*

$$\Sigma_g = \pi \left[P(h_g h_g^T) - (Ph_g)(Ph_g)^T \right]$$

after the usual symmetrization. Suspension by the roof gives the physical-time rough limit, including the deterministic area anomaly and the terminal flight residual.

Proof. Theorem 1.1 makes the Poisson series converge in the weighted Lipschitz norm. Conditional centering of D_{k+1} is exact. The same contraction applied to the bracket observable gives locally uniform predictable-bracket convergence. The drift bound yields the conditional Lindeberg estimate and uniform control of the coboundary. Martingale rough invariance then gives the first and second levels; the antisymmetric mean of the coboundary loop is the area anomaly. Joint collision/roof exponential moments and the stopped physical-clock theorem of A3 give time inversion and control of the terminal residual. $\square$

1.3. The Feynman–Kac eigenchart and nonlinear tower. For a bounded weighted Lipschitz potential V put $P_V f = P(e^V f)$.

Theorem 1.3 (Feynman–Kac eigenchart and Doob nonlinear semigroup). *There is a complex neighborhood of 0 in Lip_W on which P_V has a simple eigenvalue e^{λ_V} , a positive eigenfunction h_V , and a positive eigenfunctional ℓ_V , all analytic in V . The kernel*

$$P_V^D f = e^{-\lambda_V} h_V^{-1} P_V(h_V f)$$

is Markov. Its continuous-time suspension defines the exact nonlinear tower

$$\mathcal{E}_{s,t}^V F = \log P_{V,s,t}^D(e^F), \quad \mathcal{E}_{r,t}^V = \mathcal{E}_{r,s}^V \mathcal{E}_{s,t}^V.$$

Proof. The weighted Lipschitz spectral gap of Theorem 1.1 is stable under sufficiently small bounded multipliers. Analytic Riesz perturbation gives the eigen-data and positivity follows from the cone of positive functions. The displayed normalization sends 1 to 1. The linear Markov tower becomes the nonlinear tower after exponentiation and logarithm. $\square$

1.4. Continuous-time renewal resolvents. Let $\mathcal{L}_{V,z}$ be the roof-twisted returned operator of A2 and let E_z and X_z be the entrance and terminal-flight operators. The Laplace transform of the Doob suspension satisfies the exact renewal identity

$$(z - L_V^D)^{-1} = X_z + E_z(I - \mathcal{L}_{V,z})^{-1} E_z^*.$$

Theorem 1.4 (Integrable renewal-resolvent strip). *On a common strip $\Re z > -\gamma_0$, the suspension resolvent is meromorphic. After subtracting its finite principal parts in a smaller closed strip, the residual $R_{\text{res}}(z)$ satisfies, for $j = 0, 1, 2$,*

$$\|\partial_y^j R_{\text{res}}(c + iy)\| \leq C_j(1 + |y|)^{-1-\delta-j}, \quad -\gamma < c \leq c_0.$$

The estimate is uniform in a compact Feynman–Kac source chart.

Proof. The exact renewal decomposition separates the entrance and residual-flight integrals from the returned resolvent. The four-range Fourier theorem of A2 gives integrable bounds for $(I - \mathcal{L}_{V,c+iy})^{-1}$; differentiating E_z and X_z gains the displayed vertical powers. Analytic Fredholm theory gives finitely many poles in a compact part of the strip, while the very-high-frequency estimate gives invertibility outside that compact part. Subtract

their complete Laurent principal parts. Product differentiation of the renewal identity gives the three residual bounds. $\square$

1.5. Compressed memory with all zero multiplicities. Let R be a finite-rank orthogonal projection on a resolved Hilbert space for one Doob phase and set

$$C_V(z) = R(z - L_V^D)^{-1}R.$$

Every pole and every zero of $\det C_V$ in $\Re z \geq -\gamma$ is listed with algebraic multiplicity. For a zero ζ of order m , introduce descriptor coordinates $t^j e^{\zeta t}$, $0 \leq j < m$.

Theorem 1.5 (Exact instantaneous–descriptor–memory decomposition). *After adjoining all pole and zero Jordan chains in the strip, the compressed transfer admits the unique normalized decomposition*

$$\widehat{K}_V(z) = zR - RL_V^D R - C_V(z)^{-1} = P_{\text{inst}}(z) + P_{\text{desc}}(z) + \widehat{K}_{\text{reg}}(z),$$

where P_{inst} is the polynomial large- z part, P_{desc} is the full sum of Laurent principal parts at poles and transmission zeros, and

$$\|\partial_y^j \widehat{K}_{\text{reg}}(c + iy)\| \leq C_j(1 + |y|)^{-1-\delta-j}.$$

Consequently K_{reg} is an ordinary causal function satisfying $\|K_{\text{reg}}(t)\| \leq Ce^{-\gamma t}$. Descriptor exponents are retained with their actual real parts; no unproved minimum-phase assertion is made.

Proof. Theorem 1.4 makes C_V a finite-dimensional meromorphic matrix with high-frequency invertibility. Smith–McMillan factorization on the compact strip records every pole and zero with its full multiplicity. Subtracting all principal parts and the polynomial large- z terms gives the residual vertical bounds. Two integrations by parts on the shifted Bromwich line make the inverse transform absolutely convergent and yield exponential decay. Requiring each Jordan chain to be in monic companion form fixes the normalization and hence uniqueness. $\square$

Theorem 1.6 (Round-thirteen A4 closure). *Prepared Sinai histories possess a weighted-Wasserstein spectral gap, a correctly centered quenched rough tangent, an eigenfunction-normalized nonlinear Doob tower, and an exact instantaneous–descriptor–decaying memory decomposition. Complete-past total-variation minorization is never used.*

Proof. Combine Theorems 1.1, 1.2, 1.3, 1.4, and 1.5. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and

build gates are internal reproducibility checks and do not replace independent external mathematical review.

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